

习题12.1 p139

2(任选3道),3,6,其余剩下的任选一道大题

习题12.2 p143

26道题目中任选12道大题, 其中6道奇数, 6道偶数习题

12.3 p166

1,2,6,9,20, 其余剩下的任选5道大题(共10道题)

12.1

12.1.2

Determine the convergent points:

$$(1) \sum \frac{n-1}{n+1} \left(\frac{x}{3x+1} \right)^n: x \in \left(-\infty, -\frac{1}{2} \right) \cup \left[-\frac{1}{4}, \infty \right).$$

$$(2) \sum n e^{-nx}: x > 0.$$

$$(3) \sum \left(\frac{x(x+n)}{n} \right)^n: x \in (-1, 1).$$

12.1.3

Suppose $\sum u_n(x)$ converges to $S(x)$ on $[a, b]$. If $u_n(x)$ are non-negative continuous functions, prove that $S(x)$ reaches its minimum. Must $S(x)$ reach its maximum?

Proof: The supremum of continuous functions is a lower semi continuous function, hence it reaches its minimum (by basic topology).

However, let $u_n(x) = x^n - x^{n+1}$ then $S(x) = \begin{cases} x, & x \in [0, 1) \\ 0, & x = 1 \end{cases}$ does not reach its maximum.

12.1.6

Suppose $a > 0$, f_1 is integrable on $[0, a]$. Define $\{f_n\}$ as $f_{n+1}(x) = \int_0^x f_n(t) dt$, then $\{f_n\}$ converges uniformly to 0.

Proof: Suppose $|f_1| \leq M$, then $|f_2(x)| \leq Mx$, $|f_3(x)| \leq M \frac{x^2}{2}$, so by induction we have $|f_{n+1}(x)| \leq M \frac{x^n}{n!} \leq M \frac{a^n}{n!}$. Hence $\{f_n\} \rightarrow 0$ uniformly.

12.1.7

Suppose $\{f_n\}$ converge uniformly to $f(x)$ on I . Prove that if f_i is uniformly continuous on I , then f is uniformly continuous.

Proof: For any $x, y \in I$,

$$|f(x) - f(y)| \leq |f(x) - f_n(x)| + |f(y) - f_n(y)| + |f_n(x) - f_n(y)|.$$

For any $\varepsilon > 0$, take n such that $\|f - f_n\| < \frac{\varepsilon}{3}$. Take δ such that $|x - y| < \delta \implies |f_n(x) - f_n(y)| < \varepsilon/3$, then $|f(x) - f(y)| < \varepsilon$. Therefore f is uniformly continuous.

12.2

12.2.6

Suppose $\sum a_n$ converges, prove that $\sum a_n e^{-nx}$ converges uniformly for $0 \leq x < \infty$.

Proof: Trivial by Abel's test.

12.2.13

Prove that $\sum_{n=0}^{\infty} (-1)^n x^n (1-x)$ converges absolutely and uniformly, but $\sum_{n=0}^{\infty} x^n (1-x)$ does not converge uniformly.

Proof: $\sum_{n=0}^N (-1)^n x^n (1-x) = (1-x) \frac{1-(-x)^{N+1}}{1+x} \rightarrow \frac{1-x}{1+x}$. The difference $(-x)^{N+1} \cdot \frac{1-x}{1+x}$ tends to 0 uniformly.

However, $\sum_{n=0}^N x^n (1-x) = (1-x) \frac{1-x^{N+1}}{1-x} = 1-x^{N+1} \rightarrow 1_{[0,1]}$ is not continuous, so it does not converge uniformly.

12.2.14

Define $u_n(x) = \frac{1}{n} 1_{\{1/n\}}$, prove that $\sum_{n=1}^{\infty} u_n(x)$ converges uniformly on $[0, 1]$.

Proof: $\sum_{n=1}^{\infty} u_n(x) \rightarrow id \cdot 1_{\{1, 1/2, \dots\}}$. For any $\varepsilon = \frac{1}{N} > 0$, $\sum_{n=N+1}^{\infty} u_n(x) < \varepsilon \forall x \in [0, 1]$. Hence it converges uniformly.

12.2.15

Suppose $\{u_n(x)\} \in C([a, b])$, if $\sum u_n(x)$ converges in $x \in [a, b]$ but $\sum_{n=1}^{\infty} u_n(b)$ diverges, prove that $\sum u_n(x)$ does not converge uniformly.

Proof: Otherwise $f(x) = \sum u_n(x)$ is a continuous function on $[a, b]$ hence $\lim_{x \rightarrow b} f(x) = \pm\infty$, contradicting the Cauchy criterion.

12.2.18

Suppose $f_1 \in R([a, b])$, define

$$f_{n+1}(x) = \int_a^x f_n(t) dt, n = 1, 2, \dots$$

prove that $\{f_n\}$ converges to 0 uniformly on $[a, b]$.

Proof: WLOG assume $a = 0$ and suppose $|f_1| \leq M$ then we can show by induction that $|f_{n+1}(x)| \leq \frac{x^n}{n!} M$ which tends to 0 uniformly.

12.2.22

Suppose $\{P_n\}$ is a sequence of polynomials, prove that if $P_n(x) \rightarrow f$ uniformly on $\mathbb{R}$, then f is a polynomial.

Proof: By the Cauchy criterion, eventually P_n differ by a constant since $\|P_n - P_m\|_{\infty} < \infty$. WLOG assume $P_n(x) = P_0(x) + c_n$, then c_n is Cauchy hence converges to a constant L , so $f = P_0(x) + L$ is a polynomial.

12.2.23

Prove that

$$\sum_{n=1}^{\infty} \frac{x^n}{1 + \dots + x^{2n-1}} \cos nx$$

converges uniformly on $(0, 1]$.

Proof: Let $f_n(x) = x^n/(1 + \dots + x^{2n-1})$ then $f_n(x) > f_{n+1}(x) \forall x \in (0, 1]$ and $f_n(x) \leq \min\{x^n, 1/2n\}$.

Hence $|\sum_{n=N}^{\infty} f_n(x) \cos nx| \leq \min\left\{\frac{x^N}{1-x}, \frac{1}{\sin \frac{x}{2}}\right\}$ so for both $x \in (0, \frac{1}{2})$ and $x \in (\frac{1}{2}, 1]$ the sum is uniformly convergent.

12.2.24

Consider $f_n(x) = \frac{1}{n}e^{-n^2x^2}$, prove that: (1) $f_n \rightarrow 0$ uniformly on $\mathbb{R}$; (2) $f'_n \rightarrow 0$ on $\mathbb{R}$, but not uniformly.

Proof: (1) $|f_n(x)| \leq \frac{1}{n} \rightarrow 0$ converges uniformly.

(2) $f'_n(x) = -2nxe^{-n^2x^2} \rightarrow 0$, while $f'_n(1/n) = -2e$ so it does not converge uniformly.

12.2.25

Construct $\{f_n\}$ such that: (1) $\{f_n\}$ converges point-wise but not uniformly; (2) f'_n converges uniformly.

Solution: Consider $f_n(x) = \frac{x}{n}$, then $f'_n(x) = \frac{1}{n}$ converges uniformly to 0, but $f_n(n) = 1$ so f_n does not converge uniformly to 0.

12.2.26

Suppose $\{b_n\}$ is positive and decreasing, prove that $\sum b_n \sin nx$ converges uniformly on $[-\pi, \pi]$ iff $\lim_{n \rightarrow \infty} nb_n = 0$.

Proof: If $\lim_{n \rightarrow \infty} nb_n = 0$, then $|\sum_{k=n}^m b_k \sin kx| \leq x \sum_{k=n}^m kb_k \leq x(m-n)\varepsilon_n$ and $|\sum_{k=m+1}^{\infty} b_k \sin kx| \leq b_{m+1}/\sin(\frac{x}{2})$ implies $\sum b_n \sin nx$ converges uniformly.

Conversely, if $\sum b_n \sin nx$ converges uniformly, let $m = 2n$ and $x = \frac{\pi}{m}$ then $\sum_{k=n}^m b_k \sin kx \geq \frac{1}{2}nb_n$ so $\lim_{n \rightarrow \infty} nb_n = 0$.

12.3

12.3.1

Suppose

$$f_n(x) = \frac{1}{n\sqrt{\cos(x/n) - \cos(1/n)}}.$$

Calculate $\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx$.

Solution: Clearly $\lim_{n \rightarrow \infty} f_n(x) = \frac{1}{\sqrt{1-x^2}}$.

Note that $f_n(x) \leq \frac{10}{\sqrt{1-x^2}}$, since for $g(x) = \frac{x^2}{10} + \cos x$, $g'(x) = \frac{x}{5} - \sin x < 0$ on $x \in (0, \pi/2)$. Therefore $|f_n(x)| \leq \frac{10}{\sqrt{1-x^2}} \in \mathcal{L}^1([0, 1])$, hence by Dominated Convergence Theorem

$$\lim_{n \rightarrow \infty} \int_0^1 f_n(x) dx = \int_0^1 \frac{1}{\sqrt{1-x^2}} dx = \frac{\pi}{2}.$$

12.3.2

Let $f(x) = \sum_{n=1}^{\infty} \frac{x^2}{1+n^2x^2}$, calculate $\lim_{x \rightarrow \infty} f(x)$.

Solution: Note that

$$\frac{\pi^2}{6} - f(x) = \sum_{n=1}^{\infty} \frac{1}{n^2(1+n^2x^2)} \leq \frac{1}{x^2} \zeta(4).$$

Hence $\lim_{x \rightarrow \infty} f(x) = \frac{\pi^2}{6}$.

12.3.6

Prove that

$$1_{\mathbb{Q}} = \lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \cos^m(2\pi n!x).$$

Proof: If $x \notin \mathbb{Q}$, then $n!x \notin \mathbb{Q}$ so $\cos(2\pi n!x) \in (-1, 1)$ and $\lim_{m \rightarrow \infty} \cos^m(2\pi n!x) = 0$. Therefore $\lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \cos^m(2\pi n!x) = 0$.

If $x \in \mathbb{Q}$, then for large n , $n!x \in \mathbb{Z}$ so $\cos(2\pi n!x) = 1$ and $\lim_{n \rightarrow \infty} \lim_{m \rightarrow \infty} \cos^m(2\pi n!x) = 1$.

12.3.9

(1) Prove that

$$I = \int_0^1 \frac{1}{x^x} dx = \sum_{n=1}^{\infty} \frac{1}{n^n}.$$

(2) Obtain a similar formula for $I_1 = \int_0^1 x^x dx$.

Proof: Note that $x^x = e^{x \log x}$, so

$$\int_0^1 \frac{1}{x^x} dx = \int_0^1 \sum_{n=0}^{\infty} \frac{(-x \log x)^n}{n!} dx.$$

Here using $\Gamma(n+1) = n!$, we have

$$\int_0^1 x^n (-\log x)^n dx = \int_0^{\infty} e^{-ny} y^{n+1} dy = \frac{1}{(n+1)^{n+1}} n!$$

Therefore by Monotone Convergence theorem,

$$\int_0^1 \frac{1}{x^x} dx = \sum_{n=1}^{\infty} \frac{1}{n^n} n!.$$

(2) Analogously

$$\int_0^1 x^x dx = \int_0^1 \sum_{n=0}^{\infty} \frac{(x \log x)^n}{n!} dx = \sum_{n=1}^{\infty} \frac{1}{n^n} (-1)^n.$$

12.3.16

Suppose $f(x) = \sum_{n=1}^{\infty} \frac{\cos nx}{\sqrt{n}}$.

(1) Prove that $\int_0^{\pi} |f(x)| dx$ converges; (2) Determine $\lim_{x \rightarrow 0} f(x)$.

Solution: (1) Let $S_{n,m} = \sum_{j=n}^m \cos jx$ then

$$\sum_{n=k}^{\infty} \frac{\cos nx}{\sqrt{n}} = \sum_{n=k}^{\infty} \frac{S_{k,n} - S_{k,n-1}}{\sqrt{n}} = \sum_{n=k}^{\infty} S_{k,n} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right).$$

Since $|S_{n,m}| \leq \frac{1}{\sin x/2}$, we have $\left| \sum_{n=k}^{\infty} \frac{\cos nx}{\sqrt{n}} \right| \leq \frac{1}{\sin \frac{x}{2} \cdot \sqrt{k}}$ and clearly $\left| \sum_{n=1}^k \frac{\cos nx}{\sqrt{n}} \right| \leq 1 + \dots + k^{-1/2} \leq 2\sqrt{k}$. Hence for small x , we can take k such that $\left| \sum_{n=1}^{\infty} \frac{\cos nx}{\sqrt{n}} \right| \leq \frac{4}{\sqrt{\sin \frac{x}{2}}}$. The integral $\int_0^{\pi} \frac{dx}{\sqrt{\sin \frac{x}{2}}}$ converges, so $\int_0^{\pi} |f(x)| dx$ converges.

(2) Clearly $f(x) \rightarrow \infty$ as $x \rightarrow 0$.

12.3.17

Construct a differentiable function f such that $f(\mathbb{Q}) \subset \mathbb{Q}$ and $f'(\mathbb{Q}) \subset \mathbb{Q}^C$.

Solution: Let $\mathbb{Q} = \{q_1, q_2, \dots\}$. Consider $P_n = \prod_{j=1}^n (x - q_j)$, and $f(x) = \sum_{n=1}^{\infty} P_n(x)/m_n!$ where m_n yet to be determined. Take c_n such that $|P_n(x)| \leq c_n x^n$, $|P'_n(x)| \leq c_n x^{n-1}$ and require $m_n > c_n$, then $f(q_m) = \sum_{n=1}^m P_n(q_m)/m_n!$ is the finite sum of rational numbers, so $f(q_m) \in \mathbb{Q}$.

Let $\frac{A_{n,k}}{B_{n,k}} = P'_n(q_k) = \prod_{j \neq k, j \leq n} (q_k - q_j)$ where $A_{n,k}, B_{n,k} \in \mathbb{Z} \setminus \{0\}$, then $f'(q_k) = \sum_{n=1}^{\infty} \frac{A_{n,k}}{B_{n,k}} \frac{1}{m_n!}$. Take $N_n > \max\{|q_1|, \dots, |q_n|\}$, then $A_{n,k}, B_{n,k}, N_n$ are independent of m_n .

If $f'(q_k) \in \mathbb{Q}$ for some $k \geq 1$, then suppose $K f'(q_k) \in \mathbb{Z}$, we find $n_0 > k$ such that $0 < K \left| \sum_{n=n_0}^{\infty} \frac{A_{n,k}}{B_{n,k}} \frac{1}{m_n!} \right| < 1$ reaching contradiction. Note that

$$\sum_{n=n_0+1}^{\infty} \left| \frac{P'_n(q_k)}{m_n!} \right| \leq \sum_{n=n_0+1}^{\infty} \frac{c_n N_n^{n-1}}{m_n!}.$$

Hence it suffices to choose m_n such that $\sum_{n=n_0}^{\infty} c_n N_n^{n-1}/m_n! \rightarrow 0$ as $n_0 \rightarrow \infty$, and $\left| \frac{1}{B_{n_0,k}} \frac{1}{m_{n_0}!} \right| > \sum_{n=n_0+1}^{\infty} c_n \frac{N_n^{n-1}}{m_n!}$ for all $k < n_0$. This is clearly possible, e.g. let $m_n > c_n N_n^{n-1} + \max_{1 \leq k \leq n} |B_{n,k}|$.

12.3.18

Suppose $f \in C^\infty(\mathbb{R})$ and on any interval $[a, b]$, $f^{(n)}(x) \rightarrow \varphi(x)$ as $n \rightarrow \infty$, prove that $\varphi(x) = ce^x$.

Proof: We only need pointwise convergence $f^{(n)}(x) \rightarrow \varphi(x)$:

Since $f^{(n)}(x) \rightarrow \varphi(x)$, $|f^{(n)}(x)| \leq M(x) \forall n \geq 1$. Let $A_k = \{x \in [a, b] : M(x) \leq k\}$, then $\bigcup A_k = \mathbb{R}$. By Baire Category Theorem some A_k contains an interval I . On the interval I , $f^{(n)}(x)$ is uniformly bounded, and $f^{(n+1)}$ bounded implies $f^{(n)}$ is equicontinuous. Hence by Arzela-Ascoli $f^{(n)}(x)$ converges uniformly. Passing the limit into integrals, we obtain $\varphi(x) - \varphi(y) = \int_x^y \varphi(t) dt$. Hence $\varphi'(x) = \varphi(x)$ so $\varphi(x) = ce^x$ on I .

Assume $0 \in I$ and consider the Taylor series $\sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$, then since $f^{(n)}(0)$ is bounded, it defined a function $g \in C^\omega(\mathbb{R})$, and $f^{(n)}(x)$ uniformly bounded on I implies $f = g$ on I . Since f is analytic outside a nowhere dense set F . If $F \neq \emptyset$, take an open set $U \cap F \neq \emptyset$, then $f^{(n)}$ is uniformly bounded on U by $Ke^{|U|}$ where $K = \sup_n |f^{(n)}(a)|$ and $a \in U \setminus F$. Therefore f is analytic on U , a contradiction. Therefore f is also analytic, so $f(x) = g(x) \forall x \in \mathbb{R}$.

Note that $g^{(n)}(x) = \sum_{m=0}^{\infty} \frac{f^{(m)}(0)}{(m-n)!} x^{m-n} = \sum_{m=0}^{\infty} \frac{C}{m!} x^m + \sum_{m=0}^{\infty} \varepsilon_{n+m} \frac{x^m}{m!} \rightarrow Ce^x$ for all $x \in \mathbb{R}$, therefore $\varphi(x) = ce^x$.

12.3.19

Suppose $f(x) = \sum_{n=0}^{\infty} 2^{-n} \cos 2^n x$, determine $\lim_{x \rightarrow 0^+} \frac{f(x) - f(0)}{x}$.

Solution: The limit does not exist:

Consider $x_n = 2^{-n} \cdot 3\pi$ and $y_n = 2^{-n} \cdot \pi$, then

$$-\frac{1}{x_n} (f(x_n) - f(0)) = -\frac{2^n}{3\pi} \left(\sum_{k=0}^n 2^{-k} \left(\cos \frac{3\pi}{2^{n-k}} - 1 \right) - 2^{-n} \right) \rightarrow \frac{1}{3\pi} + \frac{1}{3\pi} \sum_{k=0}^{\infty} 2^{-k} \cos \frac{3\pi}{2^k}$$

and likewise

$$-\frac{1}{y_n} (f(y_n) - f(0)) \rightarrow \frac{1}{\pi} + \frac{1}{\pi} \sum_{k=0}^{\infty} 2^{-k} \cos \frac{\pi}{2^k}.$$

By expanding the first few terms, we know that the two limits are not the same.

12.3.20

Suppose $f(x) = \sum_{n=1}^{\infty} \frac{1}{10^n} \{10^n x\}$, prove that f is continuous and nowhere differentiable function.

Proof: Clearly f converges uniformly so f is continuous.

Suppose $x = N + \sum_{k=1}^{\infty} 10^{-k} a_k$ then

$$f(x) = \sum_{n=1}^{\infty} 10^{-n} \sum_{k=n+1}^{\infty} 10^{n-k} a_k = \sum_{k=1}^{\infty} a_k (k-1) 10^{-k}$$

Take $h_k = (10 - a_k) 10^{-k}$, $l_k = (11 - a_k) 10^{-k} \rightarrow 0$ then

$$\lim_{k \rightarrow \infty} \frac{f(x + l_k) - f(x + h_k)}{l_k - h_k} = \lim_{k \rightarrow \infty} (k-1) = \infty$$

so f is not differentiable at x .

12.3.21

Suppose $\{f_n(x)\}$ converges to a continuous function $f(x)$ on an interval I . If $\forall n$, $f_n(x)$ is differentiable on I and $\{f'_n\}$ converges to a continuous function g . It is necessary that $f' = g$ on I ?

Solution: No, consider $f_n(x) = \frac{x}{1+n^2x^2}$ then $f_n(x) \rightarrow 0$ but $f'_n(x) = \frac{1-n^2x^2}{(1+n^2x^2)^2}$ so $f'_n(x) \rightarrow 0$ for $x \neq 0$ and $f'_n(0) \rightarrow 1$, so $f' \neq g$.

12.3.22

Suppose $b_n \geq 0$, $\sum_{n=1}^{\infty} b_n \sin nx$ converges to $\varphi(x)$ on $\mathbb{R}$. Prove that $\varphi(x) \in C^1(\mathbb{R})$ iff $\sum nb_n$ converges.

Proof: If $\sum nb_n$ converges, then $\sum nb_n \cos nx$ converges uniformly on $\mathbb{R}$, so $\varphi'(x) = \sum nb_n \cos nx$ and φ' is continuous.

Conversely, if $\varphi(x) \in C^1(\mathbb{R})$, then φ is a 2π -periodic odd function, so φ' is a periodic even function. Since $\varphi(x) = \sum b_n \sin nx$ is C^1 , $\sum b_n \sin nx$ is its Fourier series, and the Fourier series of $\varphi'(x)$ is $\sum nb_n \cos nx$. Recall the Fourier series is Cesaro summable to φ' since $\varphi' \in C([0, 2\pi])$, so

$$\varphi'(0) = \lim_{N \rightarrow \infty} \sum_{n=1}^N \left(1 - \frac{n}{N+1}\right) nb_n \geq \sum_{n=1}^{\infty} nb_n.$$

Therefore $\sum nb_n$ converges (since $b_n \geq 0$).